

CMP6200/DIG6200
Individual Undergraduate Project (FYP)
Final Progress Review
Skin Care Recommendation System

Student Name: Krishu

Student ID: 23189624

Course: BSc (Hons) Computer and Data Science

Supervisor: Prakash Gautam

Reporting Period: [Date of Last Review] to [Date of this Review]

1.0 Progress to Date - Summary

The project is now in the advanced integration and testing phase, with all major system components implemented and operating together end to end. Since the last review, the data pipeline has been finalised, the multi-task diagnostic model has completed both training phases and been evaluated, the rule-based recommendation engine has been implemented and scored against the enriched product catalogue, and a small-language-model layer has been integrated to rerank and explain recommendations. The Streamlit interface and FastAPI backend are functional, and initial work on the mobile client is underway. The most significant development has been achieving strong per-class recall on the diagnostic model (88.6% overall test accuracy, 0.885 macro F1) while keeping the recommendation logic interpretable and auditable, which has reinforced the decision to keep a deterministic scoring layer between the diagnostic model and the language-model layer rather than relying on the language model alone.

2.0 Artefact Design and Development

2.1 Implementation and Current State

Three retailer datasets (ForEveryNG, Jeevee, Oriflame) have been merged into a single 14-field schema, enriched with inferred category, ingredient, skin-type and concern tags using a weighted keyword-matching approach, and deduplicated using fuzzy string matching. This produced the unified product catalogue that all downstream components query.

The diagnostic component, MultiTaskSkinModel, uses an EfficientNetB0 backbone with two task-specific heads: a 7-class condition classifier (Acne, Carcinoma, Dark Spot, Eczema, Keratosis, Milia, Rosacea) and a 3-class skin-type classifier (dry, oily, normal). Training followed a two-phase schedule: an initial frozen-backbone phase followed by fine-tuning of the top 30 layers at a reduced learning rate. On a held-out test set of 360 samples, the model reached 88.6% accuracy and 0.885 macro F1, with recall ranging from 100% for Keratosis and 96.7% for Milia down to 80% for Acne and Eczema.

The recommendation engine combines a condition-to-ingredient rule set with a composite scoring function (ingredient match, rating, review count, availability and discount), and appends a skin-type match weight when that information is available. Carcinoma predictions are excluded from cosmetic scoring and instead routed to a medical disclaimer. A Qwen2.5-based language-model layer sits on top of this deterministic shortlist, reranking and explaining the top candidates in natural language while being restricted to products already present in the candidate list, so it cannot introduce unverified recommendations.

These components are exposed through a Streamlit prototype UI for quick demonstration, and a FastAPI backend (image analysis, product listing and condition endpoints) intended as the interface for the in-progress React Native mobile client.

2.2 Analysis and Reflection

Initial testing shows the diagnostic model performs strongly on visually distinctive conditions such as Keratosis and Milia, while conditions with more overlapping visual presentation, such as Acne and Eczema, show comparatively lower recall. This is broadly consistent with expectations going into the project, but the gap was narrower than anticipated after the second training phase, suggesting the class-weighted, mixed-task sampling strategy is helping the model generalise across both tasks rather than favouring one.

These results support the project's original premise, that a purpose-trained multi-task model could reliably support product recommendation for Nepal's local retail market, while also showing that the diagnostic output alone is not sufficient. Ingredient and availability data from local retailers proved noisier than expected, which is what motivated the enrichment and confidence-scoring step; without it, category and ingredient fields were too sparse to support reliable scoring.

One specific point where the approach was revised was in how the language-model layer is used. It was originally intended to generate recommendations more directly from the diagnostic output. During development, unconstrained generation occasionally produced plausible-sounding but unavailable or inconsistent product suggestions. The design was changed so the language model only reranks and explains a candidate list already produced by the deterministic scoring engine, which removed this failure mode while keeping the more natural, explanatory output that the language model provides.

Overall, the practical work has sharpened the understanding that, for a decision affecting skin health, transparency and traceability of a recommendation matter as much as raw predictive accuracy, which has shaped the remaining design and evaluation priorities described below.

3.0 Updated Planning for Project Completion

The following high-priority tasks remain to finalise the project:

- Complete user acceptance testing of the Streamlit and mobile interfaces with a representative group of users.
- Finalise and test the mobile client against the FastAPI backend, including the image-upload and results flow.
- Run a full evaluation of the SLM reranking layer against the deterministic ranking to quantify any change in recommendation quality.
- Analyse and visualise the final evaluation results (confusion matrices, per-class metrics) for the Results chapter.
- Draft the Discussion and Conclusion chapters of the final report, including limitations around dataset coverage and medical-condition handling.
- Carry out a final proofread and formatting pass on the complete dissertation.

3.2 Draft Outline of Final Report

- Chapter 1: Introduction - problem context, aims and objectives, scope
- Chapter 2: Literature Review - skin-condition classification, product recommendation systems, use of language models in recommendation
- Chapter 3: Methodology - data pipeline design, multi-task model architecture, recommendation scoring design, SLM integration approach
- Chapter 4: Implementation - system architecture, key components, technical decisions and trade-offs
- Chapter 5: Results and Evaluation - model performance, recommendation engine evaluation, SLM reranking evaluation

- Chapter 6: Discussion - interpretation of results, limitations, ethical considerations for medical-condition flagging
- Chapter 7: Conclusion and Future Work